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Title: Stroke in Neonates and Children

Hyperacute Arterial Ischemic Stroke pathway

Stroke Screening Questions: *Was there abrupt onset of:*

F ace Asymmetrical smile?, Vision loss?, Double vision?)

A rm & Leg (Unilateral weakness or numbness? Difficulty walking?)

S peech (Difficulty speaking or understanding?, Speech slurred?)

T ime Did the symptoms have abrupt onset or sudden worsening?)

Was the answer 'yes' to any of the above questions?

Acute Arterial Ischemic Stroke Neuroprotective Care

- **NPO**
- **Head of bed flat** (if tolerated and no signs of increased ICP)
- **Normotension**: Target SBP 50-95%ile for age to maintain cerebral perfusion pressure. Tx hypotension (< 50%ile for age): NS, inotropes. Tx hypertension (> 33% above 95%ile for age): labetalol: lower by ~25% over 24 hours or rapidly if r-tPA candidate
- **Normovolemia:** maintenance NS, bolus PRN
- **Normothermia:** Treat >37C with antipyretics +/- cooling
- **Normoglycemia**: No IV glucose unless hypoglycemic, target: 5-10 mmol/L
- **Seizure control:** ASAPwith any suspected seizure activity. Consider cEEG.

No → Neurology consult

Yes →

'Stroke Alert' Criteria are met

- ❑ **Page Neurology consultant on call** indicating 'stroke alert'
    - If no response in 10 minutes, page pediatric registrar on call MD
- ❑ **Initiate NEUROPROTECTIVE procedures (see box above), ECG, and BPmonitoring**
- ❑ **Confirm time last seen well**
- ❑ **Keep NPO** and ask time of last food/drink
- ❑ **Insert PIV and draw STAT labs**
    - CBC, INR, PTT, fibrinogen, electrolytes, renal /LFTs, glucose, type & scree
- ❑ **Confirm accurate body weight**

No → Neurology consult

Yes →

ER/Pediatric on-call/Neurology (working day time only)

- ❑ Complete PedNIHSS and confirm time of symptom onset
- ❑ Order CT/CTA and STAT labs requested by ED.
- ❑ Review contraindications for r-tPA

If patient considered a candidate for acute intervention, notify:

- ❑ PICU/CCRT and request a bed for urgent admission,
- ❑ Heamatology team,
- ❑ Neurointerventionalist,
- ❑ Pharmacy

Does MRI/CT show acute thromboembolic ischemic stroke AND a large vessel occlusion on MRA/CTA?

No →

Neurology consult → No →

*** NOT a candidate for Acute intervention ***

Is there a stroke on MRI/CT?

Yes →

Admission to HD/PICU
Neuroprotective pathway (table above)

Anticoagulation and Neuroprotective Treatment Protocol

Assess Candidacy for Anticoagulation

Neurology service

- ❑ Ischemic stroke on MRI or CT
- ❑ Patient does NOT meet criteria for alteplase (tPA) administration and mechanical thrombectomy
- ❑ NO contraindications to anticoagulation as identified by stroke/ neurology team

Is the patient a candidate for anticoagulation?

No →

Continue with neuromonitoring and neuroprotective care

*Consider antiplatelet in selected cases (i.e. Moyamoya)

Yes ↓

Draw Pre-treatment Laboratory Tests
(if not already complete)

- ❑ CBC, aPTT, INR, Fibrinogen, D-dimer
- ❑ "Stroke prothrombotic panel"

****** If possible, draw prothrombotic panel prior to initiating anticoagulation, but DO NOT DELAY anticoagulation in order to obtain panel****

→

Thrombophilia screen include:

Antithrombin level
Protein C level
Protein S level
Factor V Leiden mutation
Prothrombin gene mutation G20210A
Fibrinogen level
Anticardiolipin antibodies and lupus anticoagulant

Low levels of protein C,S & anti-thrombin will need to be repeated at least 6 weeks post cessation of anti-coagulation as it can be affected by acute thrombotic state and use of anti-coagulation

↓

Consult pediatric Hematology to advise anticoagulant dosing and monitoring

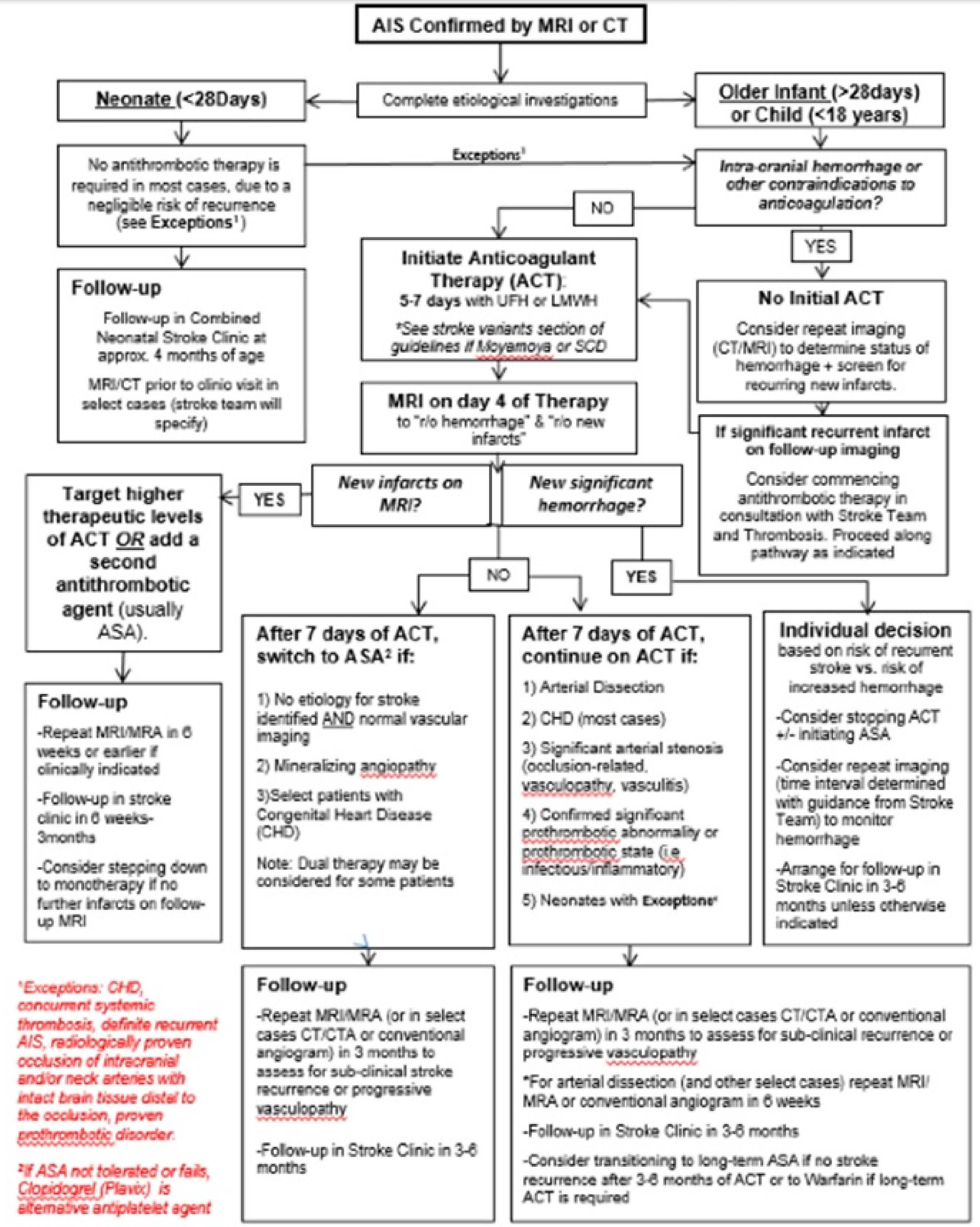
Unfractionated heparin infusion (no bolus dose) typically initial treatment choice

- ❑ Obtain aPTT and/or Anti-Factor Xa (standard heparin level) 6 hours after starting the infusion
- ❑ LMWH may be considered as initial management in selected cases

↓

Admission (if ED patient)

- ❑ ***Neuro vital signs q 4 x 24 hours unless more frequent monitoring clinically indicated, then frequency as per admitting service***
- ❑ ***Consider PICU admission if large hemispheric or posterior circulation stroke, low GCS and/or clinically unstable, seizing or in need of cEEG monitoring, or otherwise clinically indicated***
- ❑ ***NPO until feeding assessment complete***



Disclaimer: The stroke guidelines have been developed by SickKids clinicians primarily intended for the use within our institution and therefore, may need to be modified for use at other institutions. These guidelines do not indicate exclusive course of action. They are not intended to substitute for the independent professional judgment of the treating physician, as the information does not account for individual variation among patients. They are also subject to change with emerging evidence supporting new best practices in paediatric stroke.

ACUTE NEUROPROTECTIVE CARE

o **Fever:** Treat core temperature (PO/PR) > 37.5 °C with an antipyretic (acetaminophen), consider cooling in select cases in PICU.

o **Blood Glucose:** Monitor blood glucose (lab or bedside) at minimum q 6 hours for 24 hours following acute stroke, then q 12 hours for an additional 24 hours (target 5-10mmol/L). Consider treating hyper- or hypoglycemia with advice from stroke team and other consultants.

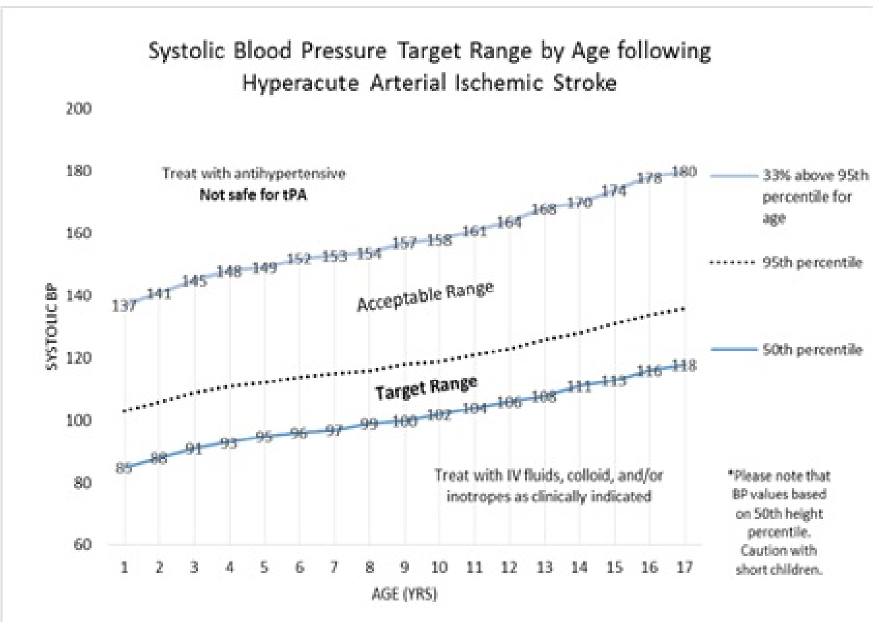
o **Blood Pressure:** In order to maintain cerebral perfusion pressure, avoid sudden reduction of blood pressure. Aim for mild hypertension targeting SBP between the 50th and 95th percentiles for age (see graph below and Appendix for age/ height specific values).

If SBP persistently below 50th percentile for age, treat with IV fluids/ colloid (or inotropes if clinically indicated). Consider antihypertensive if SBP persistently higher than 33% above 95th percentile for age.

o **Seizures:** Request spot EEG within 24 hours for all stroke patients AND with any suspected seizure activity. Mandatory cEEG monitoring for patients who present with clinical seizures or experience clear seizures within 72 hours post-stroke and in patients with large strokes with cortical involvement (i.e. involve >1/3 of the MCA territory). Consider loading dose of IV anticonvulsant (fosphenytoin/phenytoin or phenobarbital) after any suspected or confirmed seizure. Consult neurology if not already involved.

o **Fluid Balance:** Maintain normovolemia with regular monitoring of fluid balance (minimum q 6 hours).

o **Head of Bed Flat:** Keep head of the bed flat for 72 hours following acute stroke if tolerated by the patient and there are no signs of raised intracranial pressure. Otherwise, keep head of the bed at 30 degrees.



ACUTE AND PREVENTATIVE ANTITHROMBOTIC MEDICATIONS

A. **GENERAL GUIDELINES**

o **Acute Thrombolysis:** If patient 4 years of age or older, neurological deficit *began* < 6 hours from arrival to ED, and deficit is persisting, either intra-venous (within 0-4.5 hrs) or intra-arterial (4.5-6 hrs) tPA could be feasible based on joint decision by the Neurology Team, hematology, Neuroradiology, PICU and IGT**. ().**

Anticoagulant Therapy (ACT): Contact pediatric hematology for dosing and monitoring. Refer to detailed protocols for unfractionated heparin (UFH), low molecular weight heparin (LMWH), and warfarin for required baseline bloodwork (i.e. CBC, INR, aPTT, INR, Fibrinogen, D-Dimer) and laboratory monitoring guidelines.

- **Neonates with AIS:** No anticoagulant treatment is usually required due to negligible risk of recurrent stroke. Exceptions: congenital heart disease, concurrent systemic thrombosis, confirmed recurrent AIS, radiologically proven occlusion of intracranial and/or neck arteries with intact brain distal to the occlusion, proven prothrombotic disorder.
- **Older infants and children (>1 mo age) with new/acute AIS:** General practice is to anticoagulate initially for 5-7days, unless there are contraindications.
- **Monitoring during ACT:** For all children with arterial stroke who are initiated on anticoagulation, an MRI (or CT scan without contrast as second choice) should be obtained on day 4 (3-5) of therapy to assess for sub-clinical intracranial hemorrhage and to rule out further

o **Secondary preventative therapy:**

- Neonates rarely require secondary prevention therapy. Exception: neonates with BOTH a prothrombotic risk factor and echo showing persistent intracardiac defect with potential right to left shunt (i.e. persistent PFO) may be treated with ASA for prevention.
- Since older infants and children have approximately 20% risk of stroke recurrence (50% if no antithrombotic treatment, 66% if there is an arteriopathy), long-term secondary preventative therapy is needed (i.e. minimum duration of 2 years, typically indefinite). This usually consists of ASA. Clopidogrel is an alternative for single agent therapy if there is ASA intolerance or treatment failure with ASA. Circumstances may dictate prolonged anticoagulation (i.e. several months to several years) with LMWH or warfarin, such as with severe prothrombotic disorders, cardiac disease, or dissection.
- **SPECIFIC ANTITHROMBOTIC MEDICATIONS**
- Thrombolytics

o **Alteplase (tPA):** Only used in carefully defined specific stroke situations within 4.5 hours of documented stroke onset (for intravenous route) or 6 hours (for intra-arterial route) in children >4 years with a persistent, severe neurological deficit and no contra-indications. Due to major potential risks, this medication should only be given for hyperacute arterial ischemic stroke after consultation with neurology and hematology team.

Intravenous (IV) dosing:

Age < 12 years: 0.75 mg/kg (10% as bolus dose over 5 minutes, 90% as infusion over 1 hour) Maximum dose: 75 mg

Age > 12 years: 0.9 mg/kg (10% as bolus dose over 5 minutes, 90% as infusion over 1 hour) Maximum dose: 90 mg

Anticoagulants

a. **Unfractionated (standard) Heparin:** No loading dose, otherwise follow pediatric hematology dosing plan
b. **Low Molecular Weight Heparin:** Follow pediatric hematology dosing plan
c. **Warfarin:** Follow pediatric hematology dosing plan

Antiplatelets

a. **ASA:** 3-5 mg/kg/day (taken daily or 3 times/week). Round dose to nearest quarter-tablet (20 mg). Maximum dose: 325 mg/day.

Reye's syndrome is not reported at this dose. Annual influenza immunization is recommended in all children.

For Lumbar Puncture, Dental Extraction, or other invasive procedures, temporary cessation (i.e. 5-7 days) of therapy prior to procedure may be required. Consult Neurology Team and Heamatology to assess risk for recurrent stroke while off ASA for individual child. Bridging to heparin for the duration off ASA may be required.

Consider reducing ASA dose to 1-3 mg/kg/day or switching to an alternative agent if there are adverse side effects on ASA.

a. **Clopidogrel (Plavix®):** 1 mg/kg/day PO daily. Maximum dose: 75 mg/day

Clopidogrel is an alternative for single agent therapy if there is ASA intolerance or treatment failure. Note that combination therapy with multiple anti-platelet agents (i.e. ASA and Clopidogrel) is relatively contraindicated given recent evidence of increased bleeding risk in children and adults with stroke.

Stroke Guidelines - Approach to Cerebral Sinovenous Thrombosis (CSVT)

ACUTE AND MAINTENANCE ANTITHROMBOTIC MEDICATIONS

A. **GENERAL GUIDELINES**

o **Anticoagulant Therapy (ACT):** Anticoagulation is recommended in children and neonates with cerebral sinovenous thrombosis , propagation without treatment occurs in 33% of neonates and children with CSVT within one week, leading to a 40% increase in the rate of venous infarction. Anticoagulation has been shown to be safe in neonates with CSVT.

• A CT/CTV or MRI/MRV should be ordered after an interval of 4 days from diagnosis to rule out subclinical haemorrhage (if anticoagulation initiated) or to rule out propagation (if not anticoagulated). Earlier neuroimaging should be obtained as clinically indicated. Modifications for individual clinical circumstances may be necessary. Head circumference and hemoglobin level should be measured daily for neonates receiving anticoagulation.

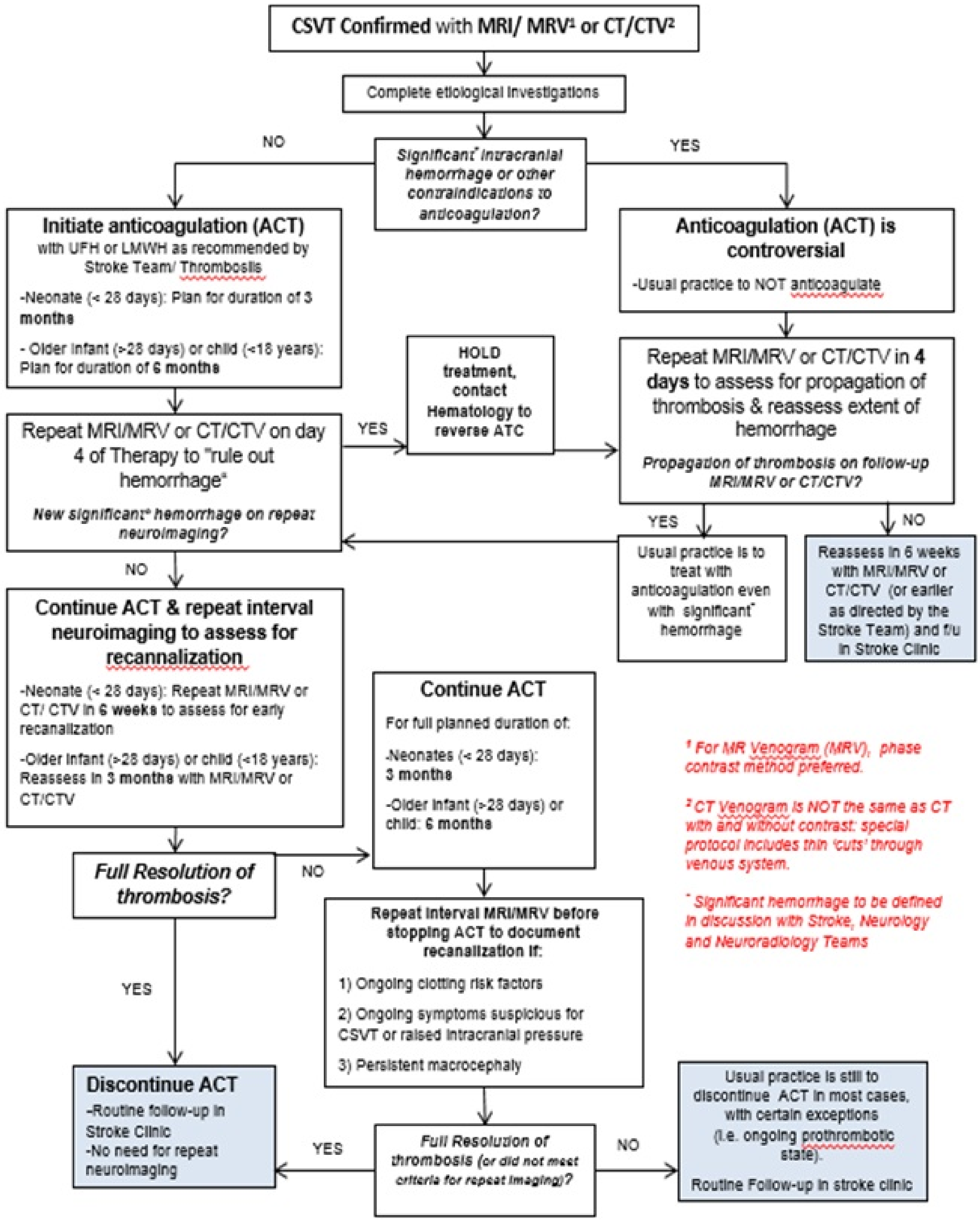
o **Unfractionated (standard) Heparin:** No loading dose, otherwise follow hematology dosing plan.

o **Low Molecular Weight Heparin:** Follow hematology dosing plan

o **Warfarin:** Follow hematology dosing plan

o **Antiplatelet agents such as ASA or clopidogrel** are generally not used in CSVT as platelets play a minor role in venous thrombosis

o **Intravascular Thrombolysis:** For progressive clinical deterioration despite anticoagulation in acute CSVT, intravascular thrombolysis may be considered.



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Hemorrhagic Stroke/Intracranial bleeding in Childhood

a. **All children presented with intracranial bleeding/ bleeding secondary to trauma or RTA, should be seen by neurosurgery service in Khawla Hospital.**
b. **In case that child presented to ER:**

- **Neuroimaging**
- **If the child has seizures treat as Status Epilepticus protocol and contact neurology team**
- **If child known to have primary bleeding disorder, consult haematology.**
- **After stabilizing the child arrange to transfer to KH.**

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